toys10_01-reflex-raja

Reflex Raja: manufacturing manual

> **Status: design only.** The three arcade games share one board (**Desi Arcade rev B**) and one firmware image; nothing has been built or tested with children.

1. What is built

1. Anyone presses a button to start a round. After a hidden random 2–6 s all four lamps light and the buzzer sounds.
2. First player to press wins a point. Pressing early loses a point and locks you out for that round. Two presses in the same millisecond is a tie (replay).
3. First to 5 points wins; the lamp of the round winner stays lit.

One board runs three games: hold a button while switching on to choose (none/button 1 = Reflex Raja, button 2 = Taal Memory, button 3 = Aloo Bomb). The blink count at power-on equals the game number.

2. Deliverables

Item	Where
PCB: Gerbers, BOM, CPL, KiCad project	`utility10/pcb/release/arcade/` (order-ready: all automatic checks pass)
Firmware for the board (8 MHz, rev B pin map)	`toys10/electronic/release/pcb/desi_arcade_rev_b.hex` (build: `make pcb` in `toys10/electronic`)
Engine + tests	`toys10/electronic/lib/toycore.c`, 80,045 host checks pass
Nano bench demo	`toys10/electronic/release/<game>.hex`

3. Bill of materials

3a. Board parts

Ref	Value	Package	Part (manufacturer)	LCSC	Assembly
U1	ATmega328P-AU	TQFP-32_7x7mm_P0.8mm	ATMEGA328P-AU	C14877	SMT (JLC)
C1	100n	C_0603_1608Metric	0603 100nF X7R	C14663	SMT (JLC)
C2	100n	C_0603_1608Metric	0603 100nF X7R	C14663	SMT (JLC)
C3	100n	C_0603_1608Metric	0603 100nF X7R	C14663	SMT (JLC)
R7	10k	R_0603_1608Metric	0603 10k 1%	C25804	SMT (JLC)
J2	ISP	PinHeader_2x03_P2.54mm_Vertical	ZHOURI 2.54-2*3	C5116479	hand/THT

Ref	Value	Package	Part (manufacturer)	LCSC	Assembly
J1	BATT 3xAAA	JST_PH_B2B-PH-K_1x02_P2.00mm_Vertical	JST B2B-PH-K-S(LF)(SN)	C131337	hand/THT
SW1	Power	SW_SPDT_PCM12	C&K PCM12SMTR	C221841	SMT (JLC)
Q2	AO3401A	SOT-23	AO3401A	C15127	SMT (JLC)
C4	10u	C_0805_2012Metric	0805 10uF X5R	C15850	SMT (JLC)
BZ1	Buzzer	Buzzer_12x9.5RM7.6	YUEXIN KC1206 16 ohm passive magnetic 12x9.5 mm (2048 Hz)	C781869	hand/THT
Q1	MMBT3904	SOT-23	MMBT3904	C20526	SMT (JLC)
D2	1N4148WS	D_SOD-323	1N4148WS	C2128	SMT (JLC)
R6	1k	R_0603_1608Metric	0603 1k 1%	C21190	SMT (JLC)
J5	Arcade button 1	JST_PH_B3B-PH-K_1x03_P2.00mm_Vertical	JST B3B-PH-K-S(LF)(SN)	C131339	hand/THT
J6	Arcade button 2	JST_PH_B3B-PH-K_1x03_P2.00mm_Vertical	JST B3B-PH-K-S(LF)(SN)	C131339	hand/THT
J7	Arcade button 3	JST_PH_B3B-PH-K_1x03_P2.00mm_Vertical	JST B3B-PH-K-S(LF)(SN)	C131339	hand/THT
J8	Arcade button 4	JST_PH_B3B-PH-K_1x03_P2.00mm_Vertical	JST B3B-PH-K-S(LF)(SN)	C131339	hand/THT
R10	330	R_0603_1608Metric	UNI-ROYAL 0603WAF3300T5E	C23138	SMT (JLC)
R11	330	R_0603_1608Metric	UNI-ROYAL 0603WAF3300T5E	C23138	SMT (JLC)
R12	330	R_0603_1608Metric	UNI-ROYAL 0603WAF3300T5E	C23138	SMT (JLC)
R13	330	R_0603_1608Metric	UNI-ROYAL 0603WAF3300T5E	C23138	SMT (JLC)
R14	100k	R_0603_1608Metric	0603 100k 1%	C25803	SMT (JLC)
R15	100k	R_0603_1608Metric	0603 100k 1%	C25803	SMT (JLC)

3b. Bought-in and mechanical (estimates, ~2,000 units)

Item	Rs
MCU + crystal	55
PCB 2-layer	30
4 arcade buttons 30 mm	60
4 LEDs + resistors	10
Buzzer	12
3×AAA holder + switch	18
Enclosure (injection, amortised)	55

Item	Rs
Packaging + manual (3 languages)	30
Assembly + test	40

- 4 arcade push buttons, 30 mm, with 5 V LED lamp (3-wire leads to the JST-PH 3-pin sockets J5-J8: button, lamp, GND), housing (ABS, or 3D-printed/laser-cut MDF for the first 500), 3 x AAA holder, button caps/labels.

4. Make the board

Order bare PCBs and SMT assembly per `manuals/STANDARD_PROCEDURES.md` A and B using the files in `utility10/pcb/release/arcade/`; solder the hand/THT parts (JST sockets J5-J8, battery JST J1, slide switch, buzzer BZ1, ISP header).

5. Program

```
avrdude -p m328p -c usbasp -U lfuse:w:0xE2:m -U hfuse:w:0xD9:m -U efuse:w:0xFD:m
avrdude -p m328p -c usbasp -U flash:w:toys10/electronic/release/pcb/desi_arcade_rev_b.hex:i
```

(8 MHz internal, brown-out 2.7 V: 3 x AAA runs from 4.5 V down to about 2.7 V.)

Pin map of the rev B board (generated from the PCB spec): buttons D0, D1, D2, A5; lamp LEDs A1-A4 (330 ohm); buzzer D9; random-seed noise on A0.

6. Bring-up

1. Batteries in, switch on: +4V5 at TP1 = 4.2-4.6 V; current under 20 mA idle.
2. Power-on blink count = selected game; hold each button in turn to select each game.
3. Each button lights its own lamp; the buzzer sounds.

7. Assemble the toy

Mount four arcade buttons in the lid, wire to J5-J8, fit the board on standoffs, screw-closed battery door, label the game-select instruction on the base.

8. Functional test: Reflex Raja (every unit)

1. Press a button to start: after a hidden 2-6 s all four lamps light and the buzzer sounds; first press wins a point.
2. Press early: that player is locked out for the round and loses a point; two simultaneous presses replay.
3. First to 5 points wins; the winner's lamp stays lit; 20 rounds: delays always between 2 and 6 s.

Game select for this game: Hold nothing (or button 1) while switching on = Reflex Raja.

Record on `manuals/templates/eol_test_sheet.md`.

9. Packaging, compliance

Toys QCO / IS 9873 applicability and testing; age grade and small-parts warnings; Legal Metrology labelling; battery safety, BIS toy-electrics, e-waste EPR, noise level; no licensed names or logos.

Add: age grade label, small-parts warning, battery warning, Legal Metrology marks; buzzer level not above 85 dB at 25 cm (verify).

10. Known limits

Host tests: false-start penalty and lock-out, locked player cannot win, tie handling, all-false-start voids the round, delay always 2.0–6.0 s with mean 4.0 s, equal players win equally often (±12 % in 4,000 simulated rounds), 800 ms debounce. Nano demo `electronic/release/reflex_raja.hex` (3.9 KB).

Button wear, 'toy electronics' certification (battery + BIS toy electrics), cheap imported reaction toys.